



Geoelectrical characterization with 1D VES/TDEM joint inversion in Urupês-SP region, Paraná Basin: Applications to hydrogeology

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ABSTRACT

Although Brazil is well known by the large rivers and the Amazon Rain Forest most cities do not have access to sufficient quantities of surface water to supply the population. Because of this 61% of Brazilian population (IBGE, 2003) depends on groundwater resources. In order to help the conscious exploration of this resource in Urupês city (São Paulo State) which is characterized by problems of lack of water, this research applied the transient electromagnetic method (TDEM) and Vertical Electrical Sounding (VES) for the geoelectrical characterization of the interest region. So, the objective of this work was increase the hydrogeological basis for groundwater exploitation of Bauru sedimentary aquifer and Serra Geral fractured aquifer (Paraná Basin). A total of 23 TDEM and 15 VES soundings were conducted during the years of 2009, 2011 and 2012. In addition, 10 pairs of VES/TDEM soundings were acquired with coincident centers to be able to perform the joint inversion. The joint inversion technique is a promising tool, which enables to get the best of both methods, where the VES add the shallow information and TDEM the deeper one. In this work, the individual and joint inversions were performed using the “Curupira” software. After data process and inversion, the results were interpreted based on geological well information provided by the Department of Water and Electrical Power (DAEE) and the Brazilian Geological Survey (CPRM) which enabled to estimate favorable places to exploitation of water in Bauru and Serra Geral aquifers. For the Bauru aquifer, the results suggest areas where thickness exceeds 100 m. In these areas, the resistivity calculated was about $20 \Omega \cdot \text{m}$. Therefore, the sediments have been interpreted as saturated sandy clay. In the basalt layer of Serra Geral Formation, the suggested locations present resistivity values $< 100 \Omega \cdot \text{m}$ at $\sim 200 \text{ m}$ depth. The indicated places in sedimentary aquifer and the locations in the fractured aquifer will may show alternative sources for groundwater exploitation and water supply for Urupês city.

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1. Introduction

In recent years, the interest in water resources in Brazil has increased, mainly by the expansion of agriculture and the growth of some cities that demand a greater amount of fresh water. In São Paulo State much of the consumed water is supplied from sedimentary aquifers (Bauru Aquifer, Guarani, São Paulo, Taubaté and others). The most important and also the more difficult to access is the Guarani aquifer,

which can reach 1300 m (top) in Presidente Prudente region. Due to the higher depths in most of its extension (Guarani aquifer), many cities choose to explore the Bauru aquifer of the Adamantina Formation (shallower). In this aquifer, the sustainable flows are around $10 \text{ m}^3/\text{h}$, with the exception of some cities near São José do Rio Preto, with sustainable flows near to $40 \text{ m}^3/\text{h}$ (Iritani and Ezaki, 2009).

In this context, Urupês city is in the Northwest of São Paulo State in the Paraná Basin. To construct wells, it is recommended geological and geophysical studies to reduce the uncertainties involved in the drilling process. Electrical and electromagnetic methods can contribute to find new points for water exploration. With these methods, it is possible to obtain an electrical resistivity model as a function of the depth which can be associated with a given geological model. This association is related to the concept that different minerals have different associated resistivity and therefore, it is possible to infer a geoelectric model in subsurface.

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In this work, the Vertical Electrical Sounding (VES) and the Time Domain Electromagnetic (TDEM) sounding were the geophysical methods used to estimate the electrical resistivity distribution in the underground. In the studied area, the VES investigates shallow depths (the first) effectively and has good sensitivity to estimate the top of shallower and resistive layers. The TDEM sounding can define well the interface between resistive and conductive layers, and has a great capacity to define deeper structures. In terms of sensitivity, the two methods are complementary. The VES defines better the shallow and resistive structures and TDEM defines well the deeper and conductive structures (Schmutz et al., 2000). Therefore, the integration of the two methods is a good option to reduce the ambiguities inherent to the results interpretation. Therefore, the integration of the two methods is a good option to reduce the ambiguities inherent to the results interpretation.

Jointly invert data sets is not a simple task, but the benefits from this technique are very promising. The results are better defined because the database is greater (Raiche et al., 1985; Bortolozzo, 2011; Bortolozzo et al., 2014a, 2014b, Bortolozzo et al., 2015a, 2015b). Some researchers such as Albouy et al. (2001) and Schmutz et al. (2000), showed how the joint inversion reduced the uncertainty of parameter determination, becoming a great tool to infer the resistivity and thickness of a geological model.

In short, this study aims to define a local insight of the geology of Urupês region through borehole information, VES and TDEM methods. With the results, it is expected to define the most promising areas for groundwater exploitation in the sedimentary aquifer (Bauru) and in the fractured aquifer (Serra Geral).

2. Geological settings

The study region is in the Paraná sedimentary basin and it is constituted mainly by three formations: Adamantina, Serra Geral and Botucatu Formations. The Adamantina formation is characterized by sand sediments with almost 90 m thickness, intercalated by clays and sand lens. Usually, this formation is associated to Bauru aquifer, which is an important water reservoir in São Paulo State. Below Adamantina formation, there is the Serra Geral formation characterized by basalt rocks. In some areas, the basalt can be fractured and filled with water (Serra Geral aquifer). According to deep wells information in the studied region, the thickness of this layer is between 400 m to 600 m (Porsani et al., 2012). The last formation is Botucatu which is formed by sandstones rocks and host the Guarani Aquifer, the most important groundwater reservoir in South America. In the Table 1 below the geological profiles of eleven wells are shown.

Table 1
Location and stratigraphy of wells in Urupês city (<http://siagasweb.cprm.gov.br/layout/index.php>).

Wells (location)	Geological description (m)
P1 – Rural road	0 to 40 – Adamantina Fm. 40 to 75 – Serra Geral Fm.
P2 – Rubens Celso Tortola Av.	0 to 60 – Adamantina Fm.
P3 – Gustavo Martins Siqueira St.	0 to 80 – Adamantina Fm.
P4 – Rural road	0 to 60 – Adamantina Fm.
P5 – Barão do Rio Branco St.	0 to 100 – Adamantina Fm. 100 to 130 – Serra Geral Fm.
P6 – Domingos Logulo St.	0 to 90 – Adamantina Fm. 90 to 152 – Serra Geral Fm.
P7 – Near to Slaughterhouse	0 to 45 – Adamantina Fm. 45 to 78 – Serra Geral Fm.
P8 – Near to Guaripú Stream	0 to 47 – Adamantina Fm. 47 to 80 – Serra Geral Fm.
P9 – Antônio Carlos Logulo St.	0 to 73 – Adamantina Fm. 73 to 151 – (crystalline rock)
P10 – City hall	0 to 104 – Adamantina Fm. 104 to 158 – Serra Geral Fm.
P11 – Madeira Stream	0 to 47 – Adamantina Fm. 47 to 74 – Serra Geral Fm.

3. VES and TDEM data acquisition and processing

Fifteen TDEM soundings and twelve VES soundings were acquired in three field campaigns conducted in 2009, 2011 and 2012. TDEM soundings were carried out with PROTEM-57-MK2 (Geonics) that allow working in three different frequencies (30 Hz, 7.5 Hz and 3.0 Hz). Data were acquired using a square central loop of 100 m side, current of 28 A. The total time for each frequency was 360 s, in this way it was possible to obtain a good signal/noise ratio.

The VES data was acquired with SYSCAL-R2 (Iris Instruments) in Schlumberger array and AB/2 length up to 200 m. All VES soundings were acquired in center of the TDEM receiver coil, to guarantee the same 1D model investigated by each VES/TDEM pair and to proceed the joint inversion. The data were statistically reduced and processed to avoid misleads in the geophysical response of each method.

The static-shift present in the VES data was corrected using the Static_Shift_Correction, a program developed in the Geophysics Department. Details about the static-shift formulations can be found in Meju (2005). Then, the data were inverted in Curupira program (Bortolozzo and Porsani, 2012), which totalizes ten joint inversions.

4. VES/TDEM joint inversion

Typically, VES and TDEM data are inverted separately, based in 1D geoelectrical models. This process, although classical, carries ambiguity when results are interpreted, because they investigate the subsurface in different ways. The ambiguity present in geophysical interpretations can normally be caused by the fact that most of the geophysical problems are poorly determined. Therefore, the parameter determination is non-unique, which means there are more than one solution for the same parameter. One way to minimize this ambiguity is to bring in additional information (or data) to the problem, which in this case can be provided by joint inversion.

The joint inversion process applied here was the same applied in Bortolozzo et al. (2014a, 2014b) and Bortolozzo et al. (2015a, 2015b). In those works are described in details how the joint inversion process works and even how the physics of the problem can weigh the contributions of both methods in the final model. Also, those articles explain the majors advantages of the joint inversion scheme used in this paper. In Bortolozzo et al. (2015a,b) the author uses the Residual Function Dispersion Maps technique (Bokhonok, 2011; Bokhonok et al., 2013; Bortolozzo and Bokhonok, 2015; Bokhonok et al., 2015) to understand the topological aspects of the objective functions of the VES, TEM and joint inversion cases. The result with this study shows how the combine objective function behaves. In a simpler and schematic form, the idea of the joint inversion can be explained by Fig. 1.

If only one method was used, the solution space could admit a large number of mathematical solutions, i.e., any of the points present in space A or B could satisfy the data set. As both methods work with the electrical resistivity property, it is natural that there is a solution space that satisfies both data sets. This space can be defined by the intersection of the spaces A and B, represented by C in Fig. 1. Note that this space restricts the number of possible solutions. The advantage of joint inversion is in reducing the space solutions allowing the parameter estimation to be more reliable. In this work, the process of VES/TDEM inversion was carried out with the “Curupira” software (Bortolozzo and Porsani, 2012) that allows the inversion of VES and TDEM data jointly and separately. Is possible to define some weight values for the soundings, but this option was not used in this research.

5. Discussion of the joint inversions results

The results presented in this section correspond to five different areas (A, B, C, D and E) as shown in Fig. 2. For each area, the most representative results of the 1D joint inversion are presented.

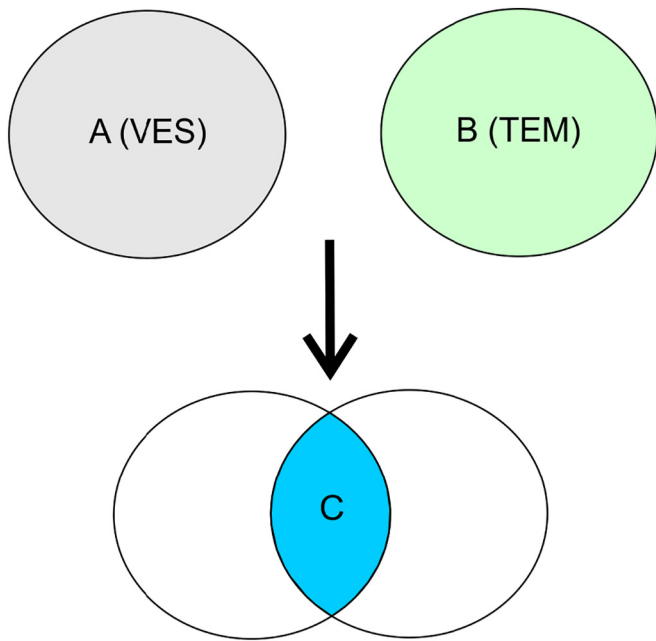


Fig. 1. Sets of solutions. Set A shows the possible solutions for a given VES data. Set B is the possible solutions to the TDEM. And C is the set of possible solutions in a joint inversion.

5.1. Area A

The VES01 and TDEM03 data used in the joint inversion 1, Fig. 3, were acquired in the northern portion of Urupês city, at an elevation of 415 m. Fig. 3 shows the results obtained by individual VES and TDEM inversions and by joint inversion. Each result (VES, TDEM and Joint) consists of data and model response curves, geoelectric model and geological profile where the results presented by inversion are interpreted. It is important to remember that not every geoelectric

layer presented is interpreted as a geological layer, in some cases, two geoelectrical layers correspond to only one geological layer.

The first result presented in Fig. 3a refers to the VES inversion. It is noted that the data misfit is 5.1% and the presented geoelectric model has 5 geoelectrical layers. Three geological layers are interpreted based on the information of the wells P4 and P8 (Table 1). According to CPRM data, the P4 well reaches a maximum depth of 60 m and its lithology is characterized by sand-clay sediments of the Adamantina Formation. The P8 well (410 m height) near P4 reaches a maximum depth of 80 m with sediments of the Adamantina Formation extending up to 47 m. From here, the basalts of the Serra Geral Formation appear.

The first geological layer in the VES01 is represented by two geoelectrical layers that are interpreted as a single layer of soil with resistivity varying from $89 \Omega \cdot \text{m}$ (1st geoelectrical layer) to $785 \Omega \cdot \text{m}$ (2nd geoelectrical layer) and thickness of 2.5 m. Fig. 4 presents a photo taken in a gully section near Urupês, where it is possible to notice the differentiation of soil colour tones; the dark part is composed by a wet and enriched with organic material and soil and the clear part is formed by drier sediments. Normally, the soil presents a great range of resistivity, since it can be composed of different sediments whose composition and granulation is poorly defined. In a short time period coarse fine to coarse sandy sediments, gravel, organic matter and other compounds can be integrated into the soil due to human occupation. In addition, the soil is also subject to weather actions that reduce or increase the amount of water in the sediment in a short period of time.

The second geological layer can be interpreted as saturated clay sandstone. It is composed by two geoelectrical layers (3rd and 4th) and, due to a very small difference in resistivity, it is interpreted as a single geological layer of resistivity $20.5 \Omega \cdot \text{m}$ and a thickness of approximately 45 m. Note that these layers could be reduced to only one, however, the data fitting is worst when performing this procedure, so we chose to include an extra geoelectrical layer and minimize the adjustment error.

The third and last geological layer can be interpreted as a layer of basalts of the Serra Geral Formation that begins at 47 m depth with resistivity of $220 \Omega \cdot \text{m}$. Note that the depth value found for the top of the

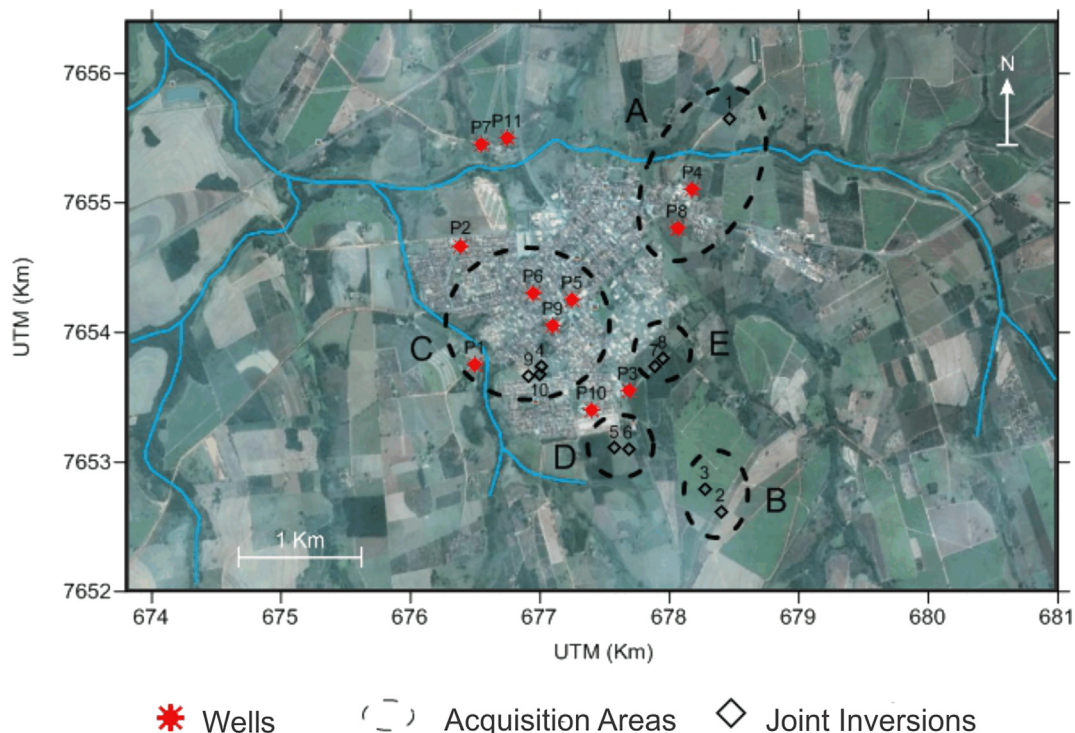


Fig. 2. Location of VES and TDEM soundings used to perform joint inversion in Urupês city. The dashed ellipses demarcate the defined areas (A, B, C, D and E).

basalt is the same as for well P8 (47 m), but different from well P4, where the sediments extend to ~60 m. However, the value of the altitude of this last well is not informed in the CPRM reports. Therefore, it cannot be taken that the different thicknesses between these two wells are due to differences in the topography or geological structures present.

Fig. 3b shows the results obtained from TDEM inversion. Note that the data misfit is 2.1% and four geoelectrical layers were estimated, which possibly define four geological layers with the following characteristics: first layer with resistivity of $65 \Omega \cdot \text{m}$ and thickness of 135 m; second layer with $412 \Omega \cdot \text{m}$ and 99 m thickness; third layer with resistivity of $80 \Omega \cdot \text{m}$ and thickness of 189 m; fourth layer with resistivity of 209 m.

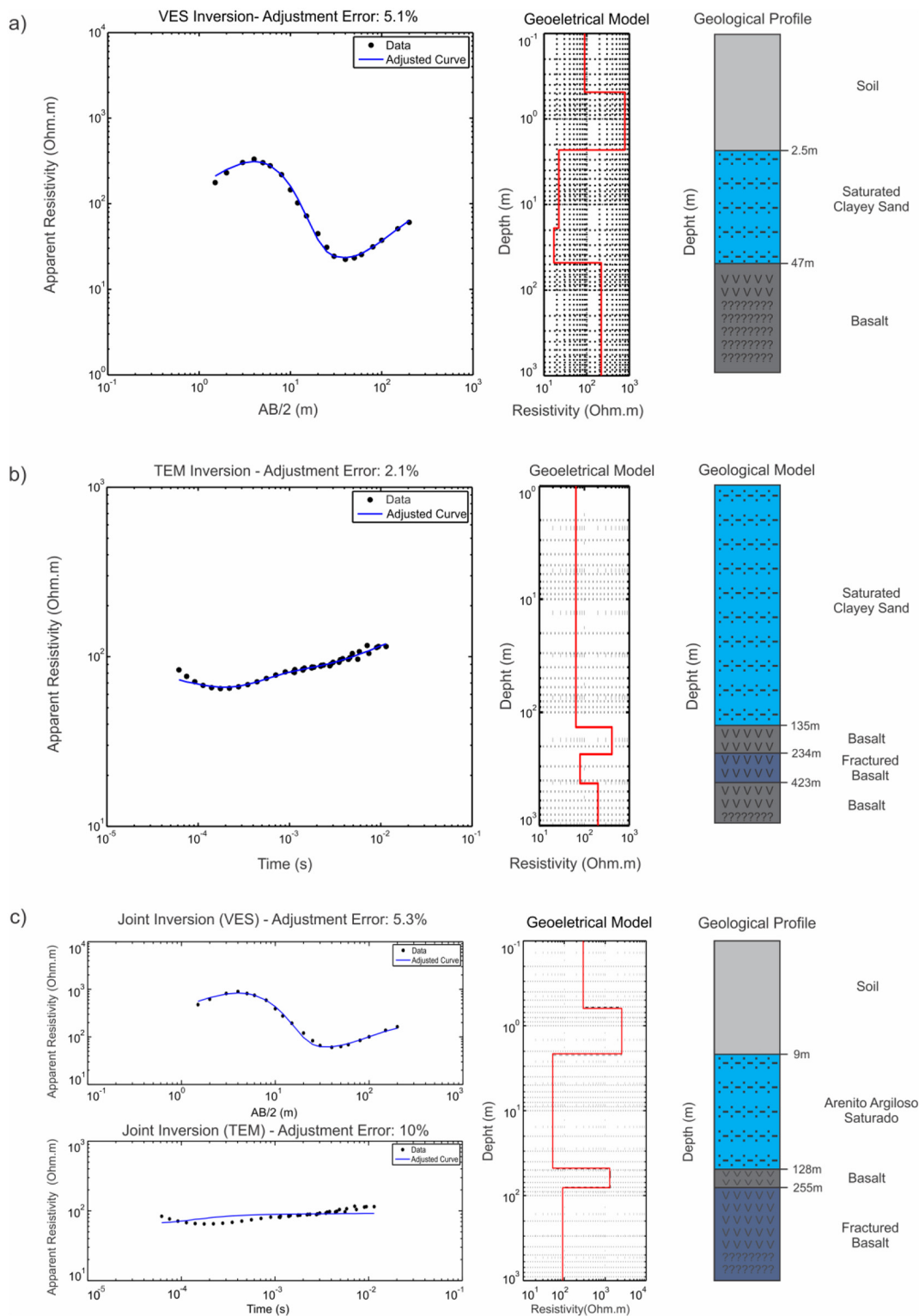


Fig. 3. Results of joint inversion 1 located in area A and the interpreted geological models on the right side. a) VES01 individual inversion. b) TDEM03 individual inversion. c) VES01 and TDEM03 joint inversion.



Fig. 4. A gully section near Urupês showing different soil tonality. The dark part is composed of a moister organic soil and drier sediments form the clear part.

Note that the first layer is relatively thick if compared to the second layer of the VES. There is a difference of 88 m between the depth of the top of the basalt in the two soundings (VES and TDEM) that cannot be explained only by the physics of the method. i.e., even if the VES and the TDEM differently solve the top of the resistive layer, the difference between the calculated depths would not be so great, since the previous modeling suggests a difference of approximately ten meters between the VES results and TDEM. Almeida (2011) also found significant differences between the TDEM results and the well data for some drilling in Bebedouro city. To explain this, it was proposed that the origin of these differences would be associated with the presence of 2D structures at the top of the basalt; when modeling the structures, it was verified that they could be the cause for the different values found (Almeida et al., 2017). Therefore, it is plausible that 2D or 3D structures are the cause of the differences between VES and TDEM for the thickness of the sediment layer of the Adamantina Formation.

The second and fourth geoelectrical layers can be interpreted as two layers of basalts of the Serra Geral Formation, because the layer resistivity values increase to $412 \Omega \cdot \text{m}$ and $209 \Omega \cdot \text{m}$, respectively. Note that the TDEM has no sensitivity to define the resistivity of very resistive structures, it identifies only as resistive layer (Fitterman, 1987; Jørgensen et al., 2005). Between these two layers, in the third geoelectrical layer, the resistivity drops to $80 \Omega \cdot \text{m}$. The decrease in resistivity value occurs at 274 m depth and therefore cannot be associated with the Guarani aquifer, since it is deeper (below 600 m). Thus, three hypotheses could explain the decreasing in the resistivity values: fractured basalt filled with water, vesicular basalt and intertrap sedimentary layers. Yamabe and Berrocal (2000) analyzed some deep wells in the interior of São Paulo State and verified that the thicknesses of the intertrap sedimentation vary from less than one meter to fifty meters thick. Therefore, this hypothesis is not reasonable, since the calculated thickness is 189 m, or almost 4 times greater than the maximum thickness found by Yamabe and Berrocal (2000). However, the presence of fractures and vesicles in basalt is common in the Paraná basin

(Etchebehere et al., 2007). In addition, sub-vertical and sub-horizontal fractures were identified during the cleaning of the P7 well in Urupês.

Fig. 3c shows the results of the joint inversion. Note that the VES is set with a 5.3% error, while the TDEM setting has a 10% error. Five geoelectric layers were defined in the initial model and their parameters were calculated, which can be interpreted with four geological layers with the following characteristics: first layer with resistivity ranging from $290 \Omega \cdot \text{m}$ to $2539 \Omega \cdot \text{m}$ and 9 m thickness; second layer with resistivity of $52 \Omega \cdot \text{m}$ and thickness of 119 m; third layer with resistivity of $1285 \Omega \cdot \text{m}$ and thickness of 127 m and fourth layer with resistivity of $95 \Omega \cdot \text{m}$. The first two geoelectric layers correspond to a single layer of soil with a thickness of 9 m. The third geoelectric layer represents the Adamantina Formation, however, the calculated thickness is distinct from that found in well P8. As discussed earlier, it is possible that sub-surface structures could cause 2D effects the TDEM data, this would explain why the TDEM curve in the joint inversion is not well adjusted. The fourth and fifth geoelectric layers correspond to the basalts of the Serra Geral Formation. The difference between them is in the values of electrical resistivity, which in the fourth layer is $1285 \Omega \cdot \text{m}$ and in the fifth layer is near $50 \Omega \cdot \text{m}$. This difference, as discussed earlier, may be related to the presence of water-filled fractures and/or vesicles within basalt.

5.2. Area B

Fig. 5 shows the results of the joint inversion 2 which is formed by the VES02 and TDEM04 data. The data were acquired in area B (southern part of the city), where two joint inversions were conducted; as they present similar results, only one of them will be discussed here. The VES02 inversion results are shown in Fig. 5a. The adjustment error is 1.9% and the geoelectric model is composed by five geoelectrical layers that can be interpreted in terms of four geological layers. The first two geoelectrical layers correspond to a single soil layer, whose calculated resistivity varies from $91 \Omega \cdot \text{m}$ to $5500 \Omega \cdot \text{m}$ with a thickness of 2.5 m. The third geoelectrical layer corresponds to the second geological layer whose resistivity value and calculated thickness is $77 \Omega \cdot \text{m}$ and 23 m respectively. This layer can be interpreted as an unsaturated sandstone layer of the Adamantina Formation. In the fourth geoelectrical layer, the calculated value for the electrical resistivity is approximately $10 \Omega \cdot \text{m}$, furthermore this layer is interpreted as a layer of clay sandstone whose thickness is 45 m (Adamantine Fm). The fifth geoelectrical layer corresponds to the fourth layer of the geological profile which is interpreted as a layer of basalt; the resistivity calculated for it is $2450 \Omega \cdot \text{m}$.

Fig. 5b shows the results of the TDEM04 inversion, in which three geoelectrical layers were sufficient to adjust the curve with an adjustment error of 1.7%. The first layer with resistivity of $1000 \Omega \cdot \text{m}$ and thickness of 39 m is interpreted as a layer of sandstone. Note that the calculated resistivity for the first layer is high ($1000 \Omega \cdot \text{m}$). As the TDEM has no resolution in the initial meters, it tends to show a layer which compositions changes from the soil later to the transition zone to the saturated portion of the sediments. In this way, the resistivity becomes an average of the resistivity of the geoelectrical layers that make up the soil and transition zone. In the second layer, the calculated resistivity is $15 \Omega \cdot \text{m}$ with a thickness of 61 m. The low resistivity value found may be associated with the presence of water and clay, so this layer is interpreted as water saturated clay-sand. Below it is the third layer, which is interpreted as a layer of basalts of the Serra Geral Formation with resistivity of $107 \Omega \cdot \text{m}$.

Note that the results for VES02 and TDEM04 show different values for the boundaries between the unsaturated and saturated portion, and for the depth of the top of the basalt. The contrasts presented between VES and TDEM results are also discussed in Raiche et al. (1985) and are due to the way each method physically behave the subsurface. While the TDEM is sensitive to conductive layers, the electrical method is sensitive to the shallower and resistive layers. Other factors such as

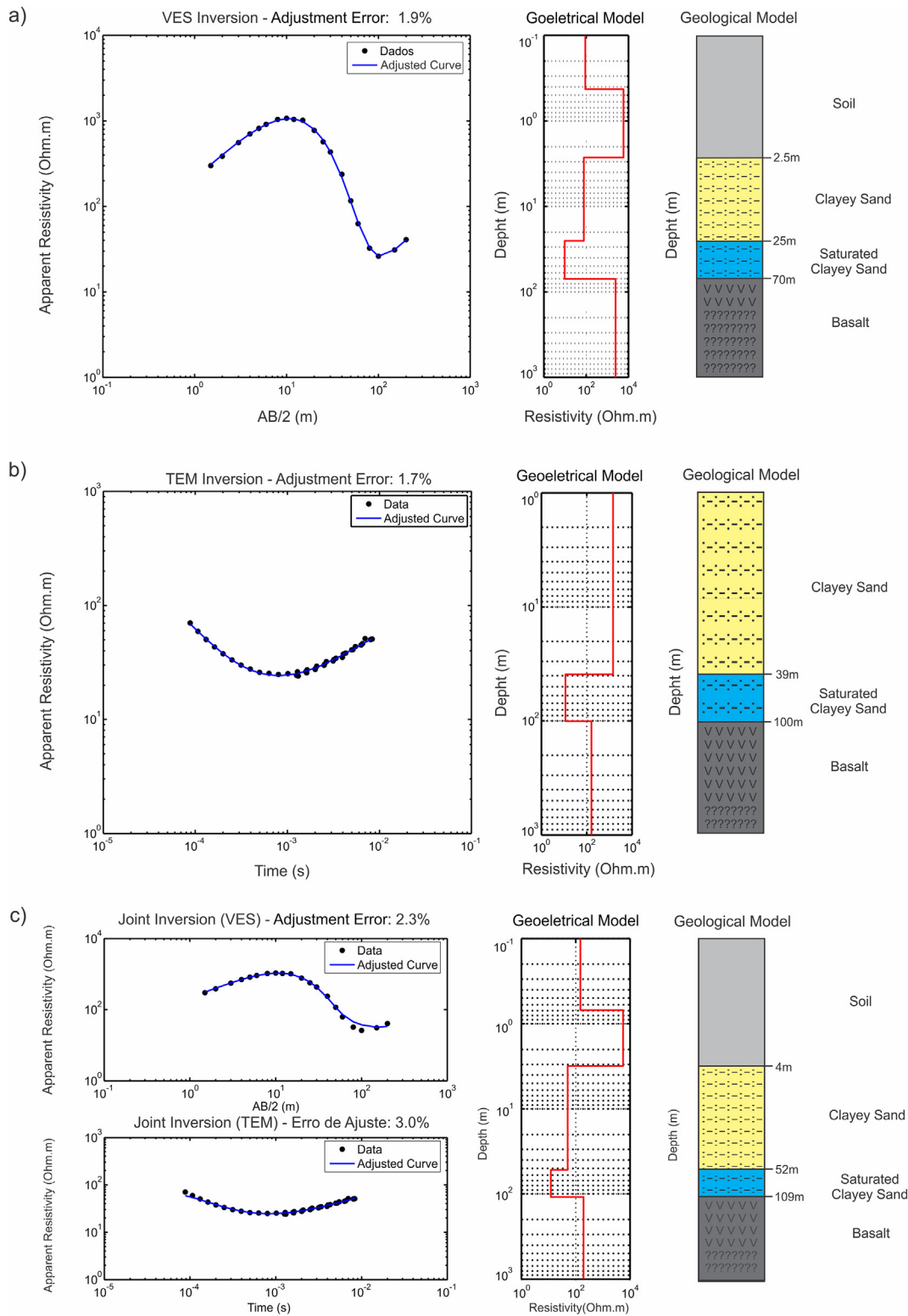


Fig. 5. Results of joint inversion 2 located in area B and the interpreted geological models on the right side, a) VES02 individual inversion. b) TDEM04 individual inversion. c) VES02 and TDEM04 joint inversion.

noise, equivalent layers and 2D structures also contribute to increase observed differences. For example, the presence of noise in a survey causes scattering of the measured points. Even if the displacement is small, it is sufficient to generate ambiguous interpretations, since the curve adjusted by VES or TDEM may present different solutions for the same dataset (Orellana, 1982).

Fig. 5c shows the results of joint inversion 2. Note that the data misfit for VES is 2.3% and for the TDEM it is 3%. Four geological layers can be interpreted in this figure. The first layer corresponds to the soil and its resistivity varies from 106 $\Omega \cdot m$ to 5500 $\Omega \cdot m$ with a thickness of 4 m. The second layer is interpreted as the unsaturated part of the sandstones of the Adamantina Formation and its resistivity is 89 $\Omega \cdot m$ with

a thickness of 48 m. The third layer corresponds to the saturated part of the Adamantina Formation, since the calculated resistivity is $9 \Omega \cdot \text{m}$ with a thickness of 57 m. The fourth layer is interpreted as basaltic with a calculated resistivity of $120 \Omega \cdot \text{m}$. Note that the model presented by the joint inversion eliminated the inherent ambiguity of an

individual interpretation by joining the two VES and TDEM databases. In this way, it was possible to estimate more accurately the thickness of the geological layers and thus to better define the range to be explored by a well in the vicinity of these surveys. In this case, the saturated portion presents thickness of 57 m and starts at 52 m depth.

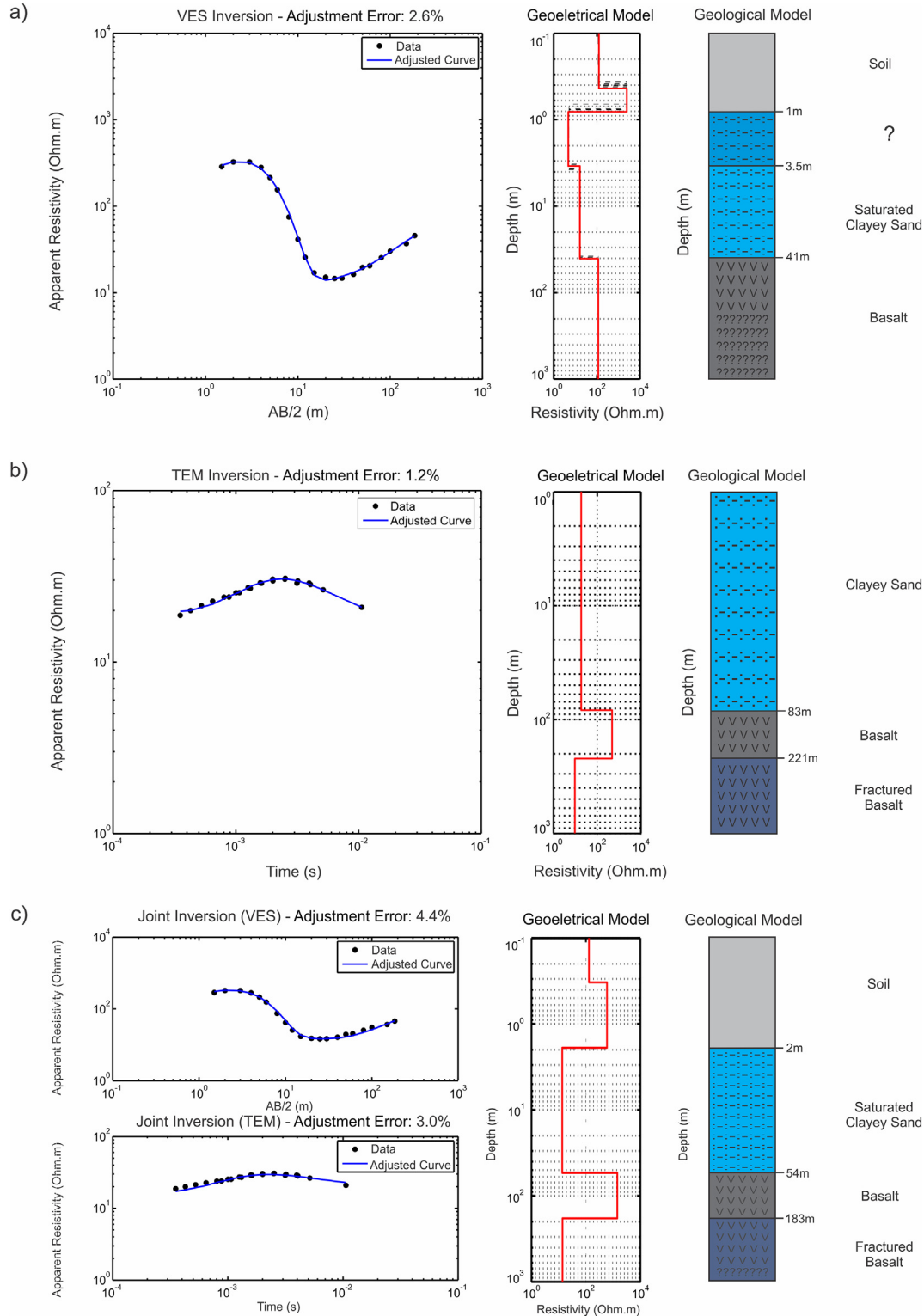


Fig. 6. Results of joint inversion 4 located in area C and the interpreted geological models on the right side. a) VES06 individual inversion. b) TDEM06 individual inversion. c) VES06 and TDEM06 joint inversion.

5.3. Area C

Fig. 6 presents the results obtained from VES06 and TDEM06 data acquired at area C (Fig. 2). This is an important area of investigation because it is inside an urban perimeter, which has a large space for VES and TDEM acquisition and, consequently allows the acquisition of data at three sites (4, 9 and 10). Around the area were located four water wells were

located, and the nearest ones are P1 and P9. The P1 well (425 m of elevation) is formed by sandy-clayey sediments of Adamantina Formation which extend until 40 m depth; from this depth starts the Serra Geral basaltic layer. Sandy-clayey sediments of Adamantina Formation and basalts of Serra Geral Formation also are present at P9 well (423 m of altitude). However, the contact between the base of sediments and the top of basalt is found at 73 m depth. This information will serve as

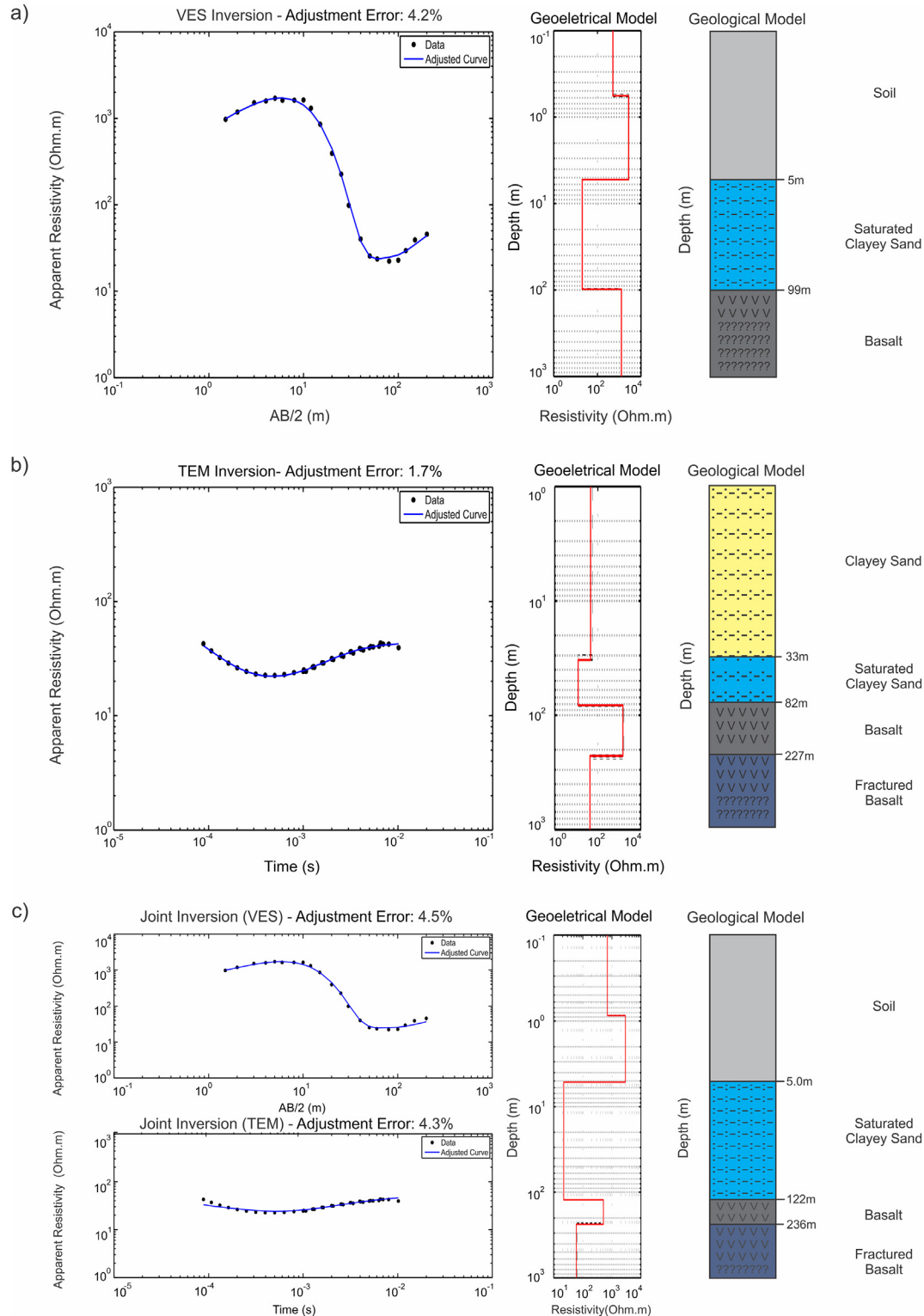


Fig. 7. Results of joint inversion 6 located in area D and the interpreted geological models on the right side. a) VES08 individual inversion. b) TDEM10 individual inversion. c) VES08 and TDEM10 joint inversion.

support to interpret the results presented in Fig. 6, since the inverted data were collected at a site situated between P1 and P9 wells.

Fig. 6a shows VES06 individual inversion results. The misfit data is 2.6% and there are five geoelectrical layers. The two first layers correspond to soil with 1 m thickness and resistivity ranges from $147 \Omega \cdot \text{m}$ to $2000 \Omega \cdot \text{m}$. The third geoelectrical layer presents $3 \Omega \cdot \text{m}$ which indicates a saturated clayey layer with 2.5 m thickness. The fourth geoelectrical layer has 38 m thickness and possibly correspond to saturated portion of clay sandstones of Adamantina Formation, because the resistivity value is $12 \Omega \cdot \text{m}$. Note that the sediments base depth, according to VES06 is very closer of value seen in P1 well. The fifth geoelectrical layer has a resistivity value of $105 \Omega \cdot \text{m}$ and can be interpreted like the top of Serra Geral basalt layer.

Fig. 6b presents TDEM06 inversion results. The data misfit is 1.2% and three geoelectrical layers were obtained. The first geoelectrical layer has electrical resistivity of $18 \Omega \cdot \text{m}$ and 83 m thickness. Observe that the value calculated of resistivity is closer to value found to VES06, which resistivity value is $12 \Omega \cdot \text{m}$. However, the 83 m thickness is different from the thickness calculated for VES06 (38 m) and in discrepancy with the value observed in well P1. Part of this discrepancy can be explained due to TDEM and VES investigates the subsurface in a different way, has mentioned in the Areas A and B discussion. The second geoelectrical layer correspond to basalt of Serra Geral Formation, the resistivity and thickness values are $481 \Omega \cdot \text{m}$ and 138 m, respectively. In third geoelectrical layer is recorded a decrease to $9 \Omega \cdot \text{m}$ in resistivity values which can be observed in geoelectrical model of Fig. 6b. As in the first case, the presence of fractures and/or vesicles filled with water could explain the values decrease of electrical resistivity.

Fig. 6c presents joint inversion results. Initially, we can note that the adjust error for VES06 and TDEM06 were 4.4% and 3%, respectively, and five geoelectrical layers were adjusted. The two first layers correspond to soil with resistivity value ranging from $129 \Omega \cdot \text{m}$ to $593 \Omega \cdot \text{m}$ and 2 m thickness. The third geoelectrical layer presents resistivity value calculated in $13 \Omega \cdot \text{m}$ and 52 m thickness and this one correspond to a saturated portion of clay sandstones of Adamantina Formation. In the fourth geoelectrical layer is recorded a resistivity increase to $1400 \Omega \cdot \text{m}$ which can be observed in geoelectrical model of Fig. 6c. Therefore, this layer can be interpreted as basaltic, extending up to 183 m depth. From this depth (183 m) starts a region with resistivity value around $50 \Omega \cdot \text{m}$; as discussed before, it can be related to fractures and/or vesicles. It must be point out that the depth of the top of the basalt in the joint inversion result is agreement with data of P1 and P9 wells, whereas the estimation of the top of the basalt would be between 40 and 73 m depth according to wells data. Besides, the fractured zone top is 183 m depth, i.e., it is more shallower than the depth found by individual inversion of TDEM06. Based on these data, it is possible to say that a water well in this neighborhood area has great possibilities to explore both aquifers: Bauru and Serra Geral; since the presence of fractures filled with water in Serra Geral Formation could increase the well flow rate.

5.4. Area D

Fig. 7 presents VES08 (Fig. 7a), TDEM10 (Fig. 7b) and inversion 6 (Fig. 7c) results. The data were acquired in area D that is composed for couple joint inversion (5 and 6) and a TDEM sounding (TDEM08) which cannot be inverted in a joint way due to low VES data quality. As the joint inversion results are similar, it will be showed only joint inversion 6 and a geological profile interpretation made with joint inversions information 5, 6 and TDEM08 (individual).

Nearby area D, there are two wells: P3 and P10. The P3 well (460 m of elevation) reaches a maximal depth of 80 m and it is composed by sandy-clayey sediments of Adamantina Formation. P10 well (455 m of elevation) has a maximum depth of 158 m with the contact between Adamantina and Serra Geral Formations around 104 m depth.

Fig. 7a shows VES08 inversion results. We can note that this inversion has data misfit of 4.2% and four geoelectrical layers. The two first layers correspond to a soil layer with resistivity ranging from $510 \Omega \cdot \text{m}$ to $2726 \Omega \cdot \text{m}$ and 5 m thickness. The third geoelectrical layer correspond to saturated part of clay sandstones belongs to Adamantina Formation with resistivity value calculated of $20 \Omega \cdot \text{m}$ and 94 m thickness. The fourth geoelectrical layer correspond to basalts of Serra Geral Formation with resistivity value calculated of $1246 \Omega \cdot \text{m}$.

In Fig. 7b we can see TDEM10 inversion results. The data misfit is 1.7% and four geoelectrical layers were adjusted. The two first layers correspond to sandy-clayey sediments of Adamantina Formation. The resistivity for the first layer, is $12 \Omega \cdot \text{m}$. This value is a consequence of the quantity of clay and water in each layer, therefore the geological profile of Fig. 7b is divided in clayey sandstone and saturated clayey sandstone. An abrupt increasing of resistivity value, $1468 \Omega \cdot \text{m}$, represents the third layer; hence it can be interpreted as a basalt layer of Serra Geral Formation. This layer extends to 227 m depth and 145 m thickness. In the fourth layer the resistivity decreases to $43 \Omega \cdot \text{m}$, indicating fractures or vesicles filled with water.

Fig. 7c presents the joint inversion results. Note that the curves are well adjusted with data misfit 4.5% for VES08 and 4.3% for TDEM10. Five geoelectrical layers were used to adjust the data, wherein two first layers correspond to soil, which thickness value is 5 m and resistivity value ranges from $695 \Omega \cdot \text{m}$ to $2929 \Omega \cdot \text{m}$. The third layer can be represented by sandy-clayey sediments of Adamantina Formation, with resistivity and thickness values of $22 \Omega \cdot \text{m}$ and 117 m, respectively. In the fourth layer, there is a resistivity value increasing to $508 \Omega \cdot \text{m}$, indicating that this layer is composed by basalts of Serra Geral Formation, and 114 m thickness. There is a resistivity decrease value of $59 \Omega \cdot \text{m}$ in the fifth geoelectrical layer. It can be associated to the presence of fractures or vesicles filled with water.

Fig. 8 shows profile 1 created with TDEM08 (T-8) and joint inversions 5 and 6 (C-5 and C-6) results. The soundings are separated by 100 m distance from each other. It is important to note that the effect of topography in this profile was not considered because the larger elevation difference between each sounding was 1 m; the GPS used to determine the altitude has an error around ± 5 m. We can note in profile 1 that the uppermost part of individual inversion (TDEM08) is different of the joint inversion results (C-5 and C-6). There is a layer relatively more resistive (clayey-sand) than others in TDEM08. Despite TDEM does not estimate well shallow resistivity and thickness layers, the effect of shallow conductive layers can be change eddy currents physical behavior in this layer, hence it can be noticed by the method. Thus, the initial layer of

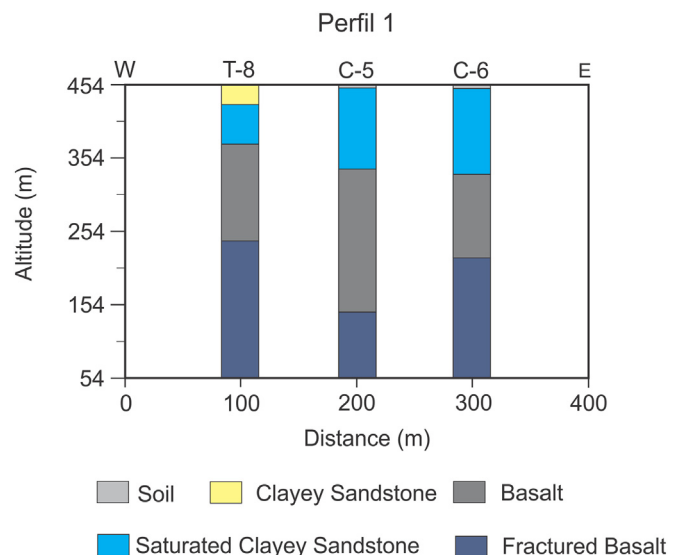


Fig. 8. Geological profile 1 in E-W direction.

TDEM would presents an average behavior between initial layers and saturated portion of sandstones clay Adamantina Formation. Besides, it is possible to note some local geology peculiarities in profile 1. Bauru aquifer thickness is variable and the basalt top can be found

around 120 m (334 m altitude) depth. Basalt thickness is very irregular being thin in T-8 (132 m) and C-6 (114 m) soundings.

Regarding the hydrological potential of the studied area, it is possible to say that the Bauru aquifer has a good potential for water exploitation,

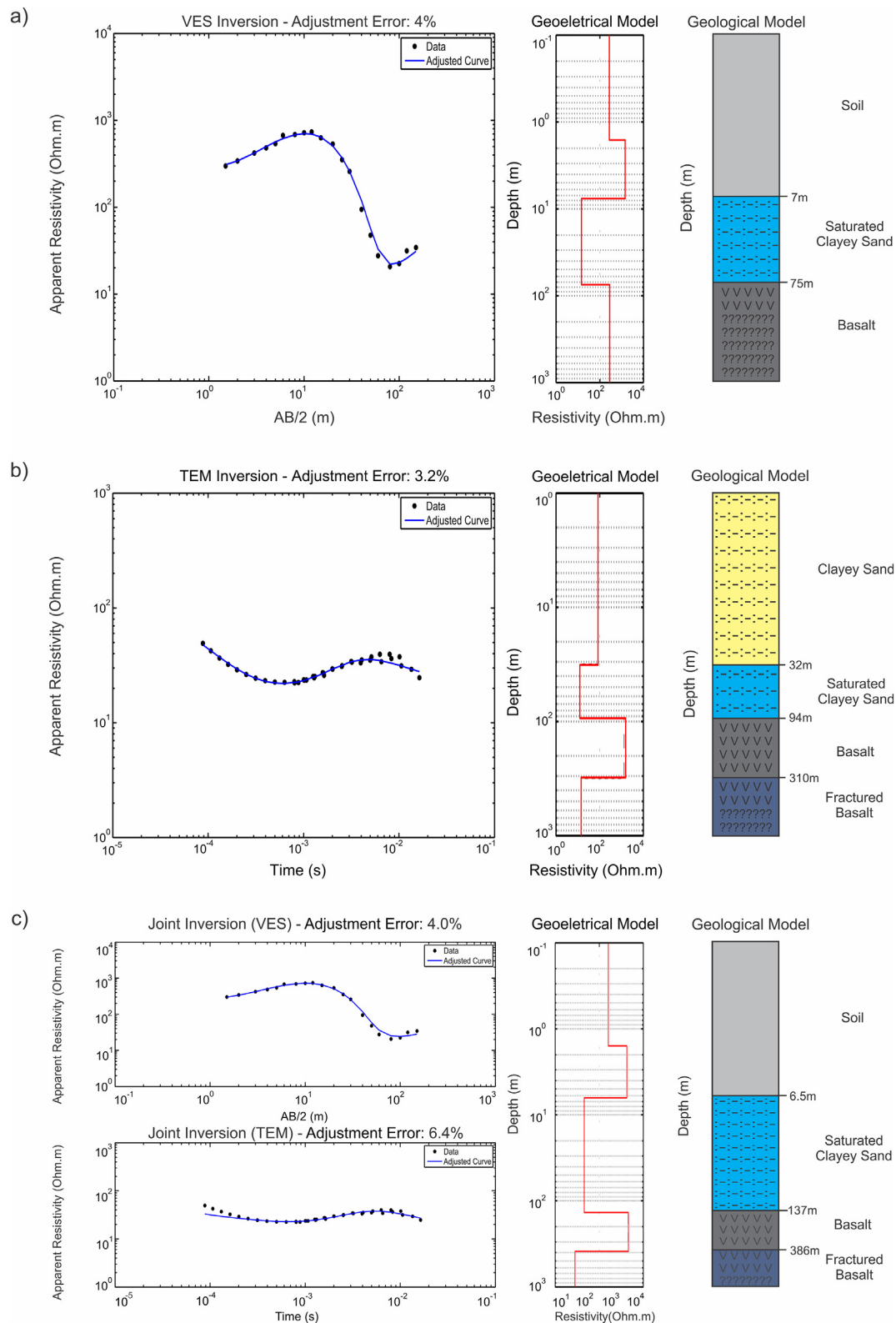


Fig. 9. Results of joint inversion 8 located in area E and the interpreted geological models on the right side. a) VES10 individual inversion. b) TDEM12 individual inversion. c) VES10 and TDEM12 joint inversion.

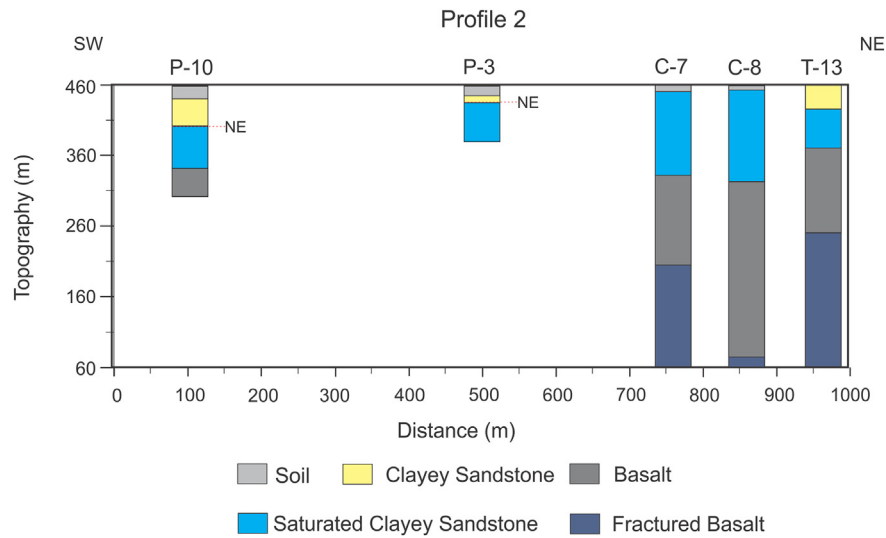


Fig. 10. Geological Profile 2 in NE-SW direction.

because the three soundings profiles (T-8, C-5 and C-6) show saturated thick regions that can be exploited. Besides, fractured or vesicle basalt also can be explored, because of low resistivity, they might be filled with water, indicating good places for the Serra Geral aquifer exploration.

5.5. Area E

Fig. 9 presents VES10 (Fig. 9a), TDEM12 (Fig. 9b) and joint inversion 8 (Fig. 9c) results from data acquired in area E, close to area D. This area is composed by a couple of joint inversion (7 and 8) and a TDEM

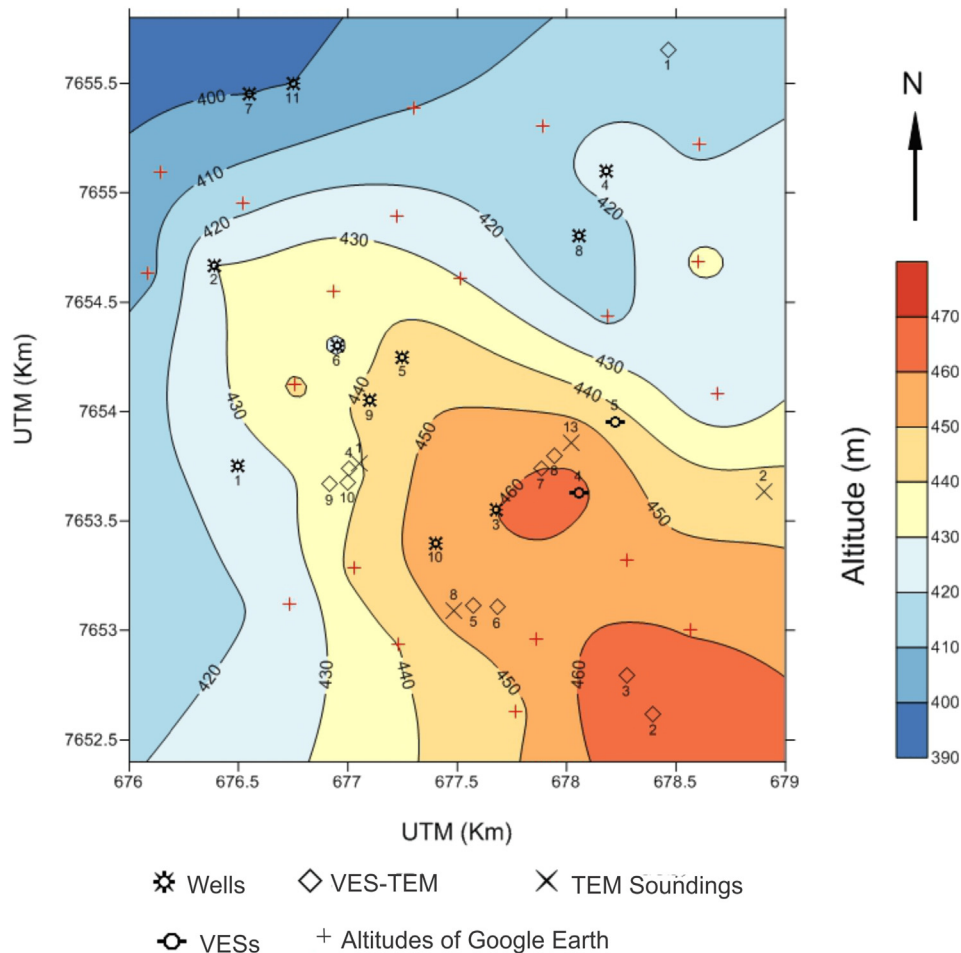


Fig. 11. Topography map of Urupês -SP.

sounding (TDEM13) which could not be jointly inverted due to VES low data quality. As the joint inversion results are very similar, it will be shown only the joint inversion 8 results and a profile made with the soundings information.

Fig. 9a shows VES10 inversion results. The data misfit is 4% and three geoelectrical layers can be interpreted. The first layer corresponds to soil and it is composed by two initial layers with 7 m thickness and resistivity ranging from $275 \Omega \cdot \text{m}$ and $1453 \Omega \cdot \text{m}$. The second layer has resistivity and thickness value of $15 \Omega \cdot \text{m}$ and 68 m, respectively, and it can be interpreted as a saturated portion of clayey sandstones of Adamantina

Formation. The third layer whose resistivity value is $287 \Omega \cdot \text{m}$, possibly corresponds to Serra Geral Formation.

Fig. 9b shows TDEM12 results. For this inversion, the data misfit is 3,2% and four geoelectrical layers were obtained with the following characteristics: layer 1, resistivity and thickness values of $83 \Omega \cdot \text{m}$ and 32 m respectively; layer 2, $12 \Omega \cdot \text{m}$ and 62 m; layer 3, $1558 \Omega \cdot \text{m}$ and 216 m; layer 4, resistivity value of $14 \Omega \cdot \text{m}$. Based on P10 and P3 wells data, it was possible to interpret the four geoelectrical layers. The first geoelectrical layer can be interpreted with a variable composition, in other words, it is possible that it is composed by soil and a transition

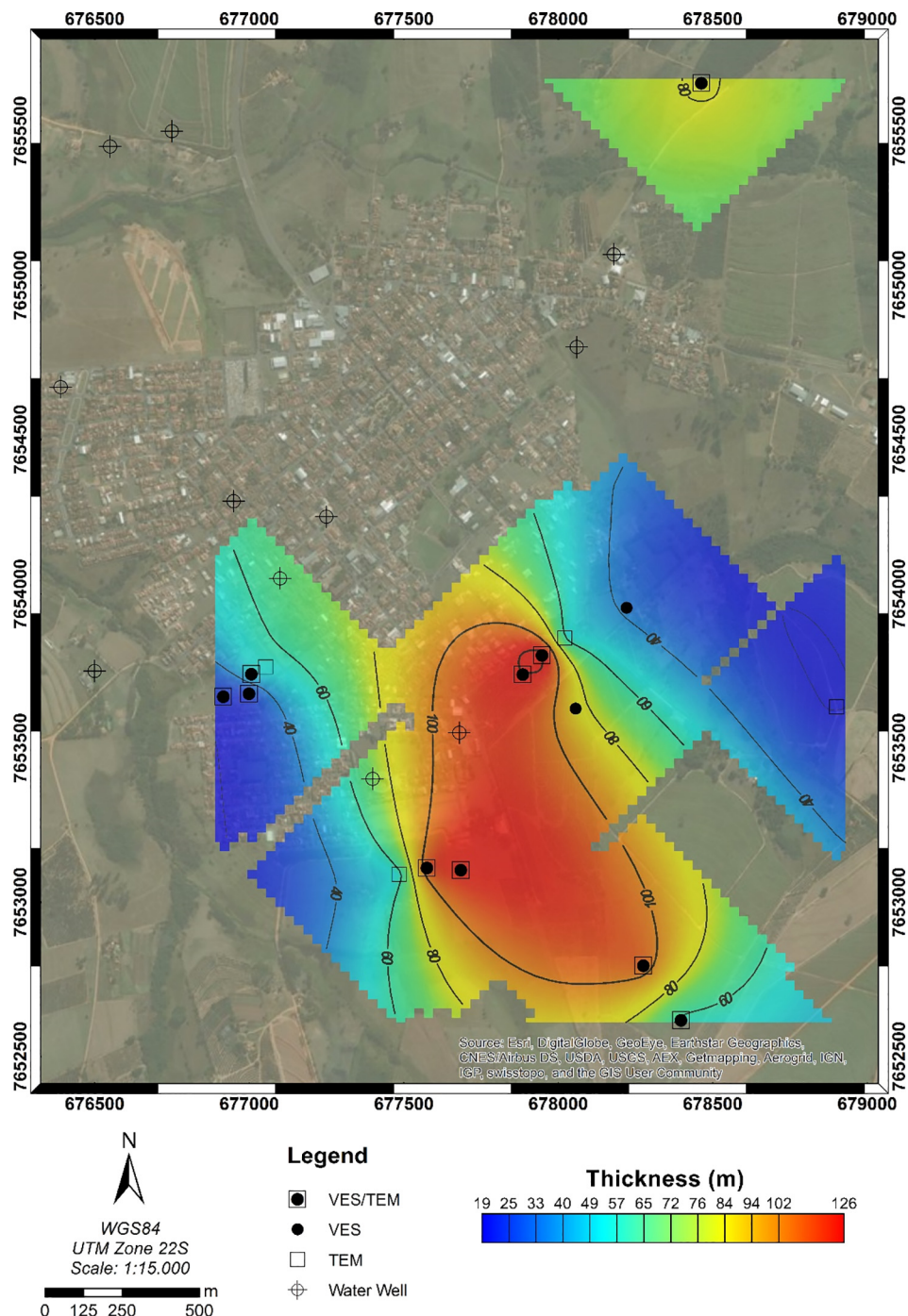


Fig. 12. Thickness map of Bauru aquifer.

zone between unsaturated and saturated clayey sandstones of Adamantina Formation. The second layer is represented by Adamantina Formation portion extending to 94 m depth. The third geoelectrical layer is represented by Serra Geral Formation basalts. In the fourth layer, there is a significant resistivity decreasing that can be observed in the geoelectrical model of Fig. 9b. As it was explained in the other soundings, this resistivity decreasing can be associated to fractures and/or vesicle filled with water.

Fig. 9c presents the joint inversion 8 results. Altogether, five geoelectrical layers were adjusted with data misfit of 4% to VES and 6.4% to TDEM. The following parameters were calculated: layer 1, resistivity and thickness values of $262 \Omega \cdot \text{m}$ and 1.5 m, respectively; layer 2, $1810 \Omega \cdot \text{m}$ and 5 m; layer 3, $20 \Omega \cdot \text{m}$ and 132 m; layer 4, $2124 \Omega \cdot \text{m}$ and

249 m; layer 5, resistivity value of $8 \Omega \cdot \text{m}$. The two first layers combined correspond to a unique layer of soil with 6.5 m thickness. The third geoelectrical layer is interpreted as a saturated portion of Adamantina Formation, extending to 137 m depth. The fourth layer, due to the high resistivity value ($1810 \Omega \cdot \text{m}$), can be interpreted as basaltic (Serra Geral Formation). The last one can be related with fractured basalts filled with water.

Fig. 10 shows all results obtained in area E in a single profile (2). The information used to make this profile comprises geological description from wells (P10 and P3), a geological profile obtained through joint inversions (C-7 and C-8) and TDEM13 individual inversion (T-13). The profile topography is plane because the larger difference between soundings and wells elevations is 5 m. As the GPS error is 5 m, this

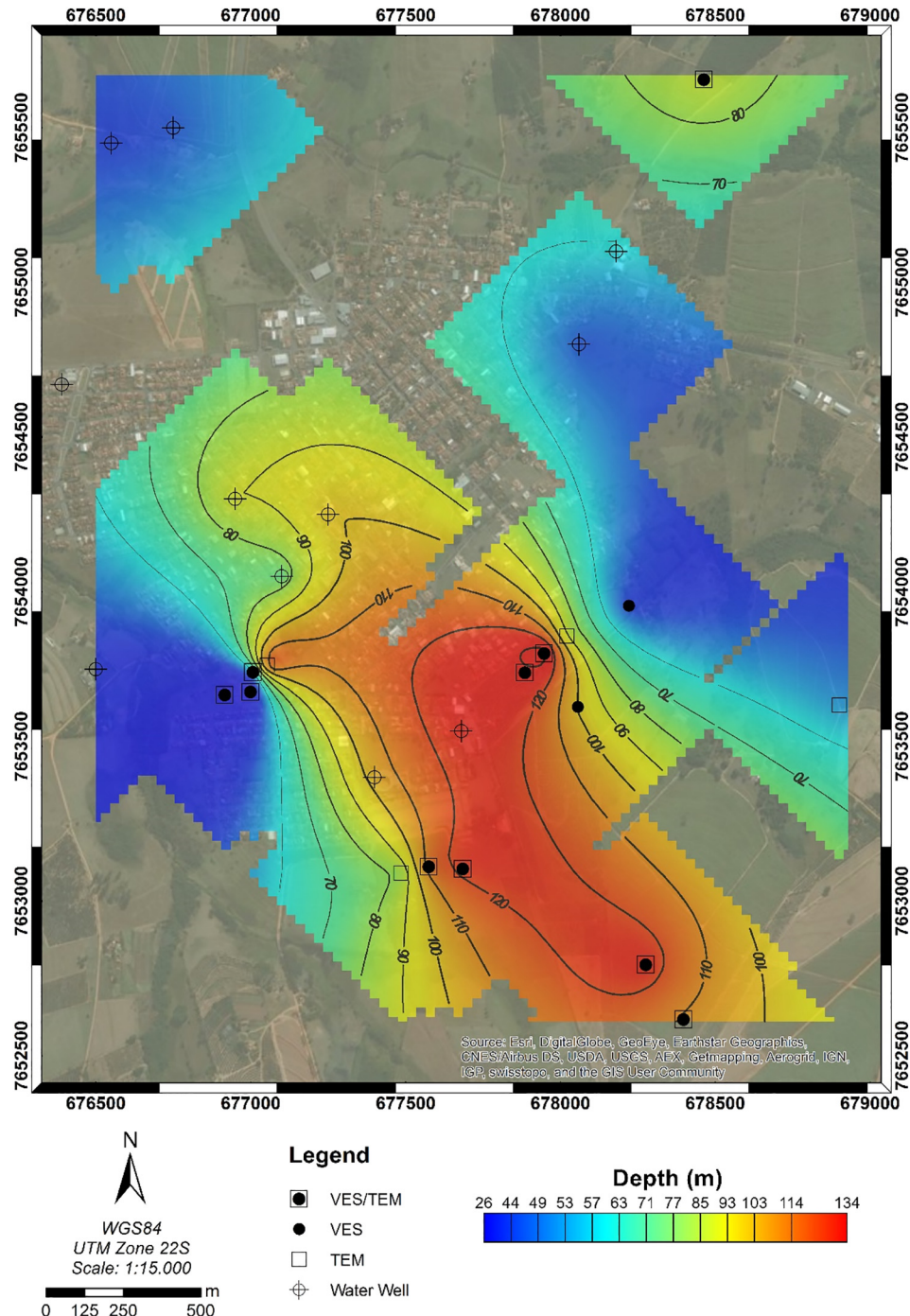


Fig. 13. Depth map of the top of basalt.

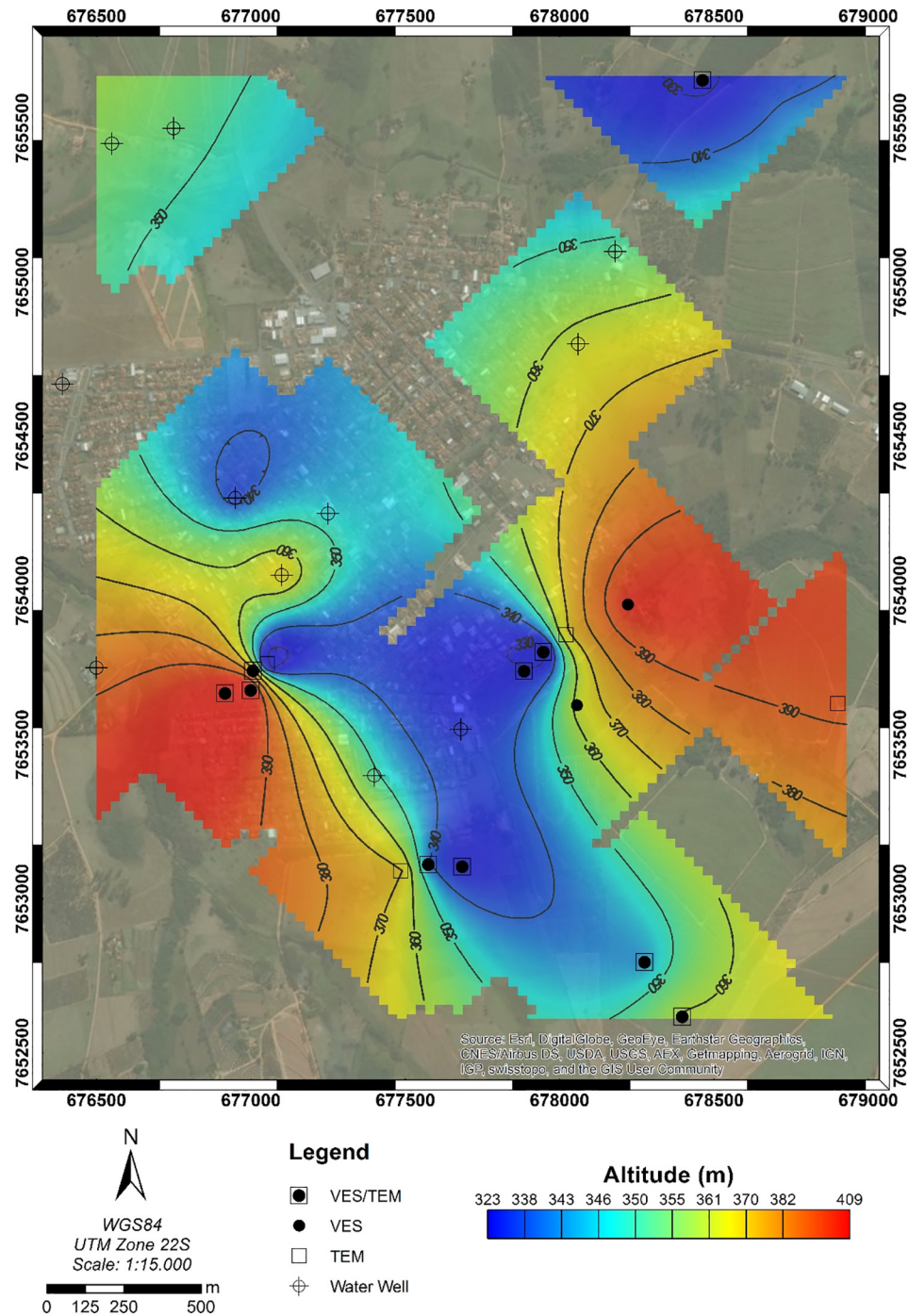


Fig. 14. Altitude map of the top of the basalt.

difference is in the accuracy interval. Thus, we admit that the topography is effectively plane. Note that basalt top depth is higher than 100 m along the profile and Bauru aquifer thickness (saturated clayey sandstone) is wide variable. The basalt fractured (or vesicle) region has 209 m depth or 251 m altitude. It is important verify that in C-8 joint inversion, the top of the fractured region is different of adjacent soundings. As it was explained before, this abrupt difference between soundings can be explained by the presence of 2D structures.

However, an exploration of the Serra Geral aquifer will have a great cost, since the fractured basalt has been found very deep (close to 250 m depth). A good estimate of the best points for drilling wells for groundwater exploration should be addressed in discussions of aquifer thickness maps.

Concerning to hydrological potential, this area has a good potential in relation to Bauru aquifer, mainly because the thickness of the saturated layer is around 100 m. The exploration of Serra Geral aquifer demands a lot of resources, once the fractured basalt is very deep (near of 250 m depth). A good estimation of the better points to drill wells for groundwater exploration is addressed in discussions of maps of Bauru aquifer thickness and Serra Geral top aquifer, which will be presented next.

5.6. Maps

The maps in this section were elaborated using the software Surfer 10 to obtain a general estimation of thickness and depth values of the Bauru and the Serra Geral aquifers. To construct the maps, we used

the results of the joint and individual inversions, wells data and Google Earth information (for altitude maps). The method chosen for data interpolation were Kriging. This interpolation method is recommended because the result obtained gives a good idea about the data tendency, especially when the interpolation mesh is irregular. The Kriging produces the best unbiased linear data estimation of a data attribute in a non-sampled location (Isaaks and Srivastava, 1989). Fig. 11 shows the topography map of Urupês. This map was created with the altitudes obtained in the field and in Google Earth. This tool was chosen to improve the database. In this map, we can see that the highest altitudes are in the central portion, near the VES/TDEM 7 and 8 soundings, and southeast of the city (VES/TDEM 2 and 3). On the other hand, the lower regions are in the north and west where the altitude varies from 400 m to 430 m.

Fig. 12 shows the thickness map of Bauru aquifer. It is observed in this map that the greatest thicknesses are coincident with the high values of the topography of the area and reaches the highest values in the central and southern portions of the city, with thicknesses close to 120 m. The lowest values of thickness are in the eastern portions, with values close to 40 m, west and north (~50 m) of Urupês city. Through the thickness map it is possible to obtain an estimation of the best drilling points in Bauru aquifer. As can be seen, the larger thicknesses coincide with the higher topography. Therefore, the adjacent areas to the VES/TDEM 2, 3, 5, 6, 7 and 8 soundings may represent excellent locations for drilling a well due to the large thickness of the Sedimentary Aquifer at these points.

Fig. 13 shows the depth map of the top of basalt. The lowest depths of the top of the basalt are in the peripheral regions of the city, north, east and west. At these points, the top of the basalt is approximately 50 m deep. The exceptions are the southern part of the city (VES/TDEM 2 and 3), where the top is found at depths between 105 m and 115 m, and the central portion. The depth of the basalt varies smoothly in the NW-SE direction. This direction coincides with the previous maps, indicating that the basalt can follow the topography. On the other hand, the depth variation is accentuated in the E-W direction. Note that in the center-east region there is a variation of approximately 80 m in the top of the basalt depth in <500 m distance, which may indicate a 2D structures.

Fig. 14 shows the altitude map of the top of the basalt that was generated subtracting the sounding point altitude by the basalt depth value. Through this map, it is possible to establish a relation with the local topography. We can observe that the top of the basalt is practically constant in the SE-NW direction. In this direction, there is a remarkable increase in the altitudes values (topography) in the SE direction, thus contrasting with the altitude values of the top of basalt that practically remain constant in the direction SE-NW direction. In this way, it is noticed that the top of the basalt does not vary with the topography in that direction. On the other hand, there is great variation of altitude of the top in the E-W direction, mainly in the east direction (between TDEM13 and VES05). This variation in a short space could be explained by the presence of structures, such as faults and fractures that are common in the Paraná Basin. These faults can be related to old tectonic efforts or to the magma cooling. According to Etchebehere et al. (2007), the main lineaments present in the region favoring water exploration correspond to the directions N40, N15 and N350. The top of the basalt is shallow near the soundings demarcated by ellipses. In the eastern and western parts of the Urupês region, the possibility of fractured basalt is greater, since TDEM soundings indicate a region with low resistivity below basalt. The northern portion is one of the regions where the top of the basalt is shallow, around 40 m deep. In the central and southern parts of the city, close to the joint inversions (VES/TDEM) 5, 6, 7 and 8, the basalts may be fractured, but it is at great depths, close to 200 m.

6. Conclusions

Generally, the VES/TDEM joint inversion proved to be an efficient tool to support the geoelectrical data interpretation, exploring better shallow structures (VES) and deeper structures (TDEM).

Concerning to aquifers characterization, it was verified that the Bauru sedimentary aquifer was well characterized. The results showed that the largest thicknesses are found in the central and southern portions of Urupês city, near to VES-TDEM 2, 3, 5 and 6 soundings. In these soundings, the calculated thicknesses values were >100 m and the electrical resistivity values ~20 $\Omega \cdot m$, indicating a sedimentary composition sand-clayey filled up with water.

The presence of some possible fractures in Serra Geral aquifer were also determined in VES-TDEM 4, 5, 6, 7 and 8 soundings and in TDEM 2, 8 and 13 soundings. In these soundings, the low values of resistivity (<100 $\Omega \cdot m$) indicates possible fractures and/or vesicles that can be filled up with water and clay. To explore this aquifer, it is recommended to drill a well near to TDEM 2, whose basalt top is around 50 m and resistivity <100 $\Omega \cdot m$, and near to VES-TDEM 4 soundings, where fractured zone ~180 m depth. These two points represent the best location for drilling a well, since the costs of drilling would be lower due to lower sediments and lower basalts thicknesses to be drilled. These results may serve as a guide to wells drilling for groundwater exploration in Urupês city.

VES-TDEM 1, 11 and 12 soundings presented anomalous behavior with significant differences between the values obtained by VES and TDEM that prevented a good adjust in the joint inversions for these soundings. The differences could be associated to 2D and/or 3D structure in subsurface, since the results obtained through the joint inversions are different of the well data located near to these soundings.

The altitude maps of basalt top and topography showed that the basalt layer do not follow the topography in SE-NW direction. While the altitude increases in NW-SE direction, the basalt top stay practically constant. In NW direction, near to wells 7 and 11, the basalt top can be found near to 50 m depth.

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